SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1*:* Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I K.Bhavya Sri/192472245 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Bhavya

Analytical Problem Solving

Intelligent Delivery Agent Using a Weighted Graph

Introduction

The delivery network is represented as a weighted graph in which nodes represent locations, edges represent connections between locations, and edge weights represent travel costs. The intelligent agent must find the best route from the starting location S to the destination G while minimizing the total cost.

1. What are the percepts for this agent?

Answer:

Percepts are the information that an agent receives from its environment to make decisions.

The delivery agent perceives the following information:

- Current location: The node where the delivery vehicle is currently present.
- Destination: The target node G.
- Available routes: Connections between nodes, such as S–A, A–B, and A–C.
- Route costs: The numerical weights assigned to each connection.
- Route availability: Whether a route can be used.
- Delivery status: Whether the package has reached its destination.

Analysis: From the given graph, the agent observes that S connects to A with a cost of 1 and directly to G with a cost of 12. It can use this information to compare possible routes rather than automatically choosing the direct route.

2. Characterize the operating environment.

Answer:

The delivery agent operates in a graph-based route-planning environment. Assuming that the graph and all edge costs are known and remain unchanged, the environment can be classified as follows.

Environment property	Classification	Justification
Observability	Fully observable	All locations, routes, and costs are known.
Determinism	Deterministic	Each route has a fixed cost and predictable destination.

Task structure	Sequential	Each movement affects the next decision and total cost.
Change over time	Static	Routes and costs remain unchanged during planning.
State space	Discrete	Locations and connections are distinct.
Number of agents	Single-agent	One delivery agent plans and executes the route.

Analysis: These properties make the problem suitable for graph-search algorithms. If traffic conditions or route costs change, the environment would become dynamic, and the agent might need to recalculate its route.

3. What are the actions the agent can take?

Answer:

The agent's actions are the operations it performs to reach the destination.

1. Move: Travel from the current node to a connected node.
2. Select a route: Choose among the available connections.
3. Calculate cost: Add the weights of the edges in a path.
4. Compare routes: Evaluate alternative paths to G.
5. Choose a path: Select the path with the lowest total cost.
6. Replan: Find another route if a connection becomes unavailable.
7. Complete delivery: Stop when the destination G is reached.

Analysis: The agent should not select a route only because it is direct. It must consider the cumulative cost of reaching the destination. For example, travelling through A and C may be cheaper than travelling directly from S to G.

4. How can one evaluate the performance of the agent?

Answer:

The agent's performance can be evaluated by measuring its ability to reach G successfully while minimizing the total route cost.

The main performance measures are:

- Total cost: Sum of the weights along the selected route.
- Goal achievement: Whether the agent reaches G.
- Route optimality: Whether the chosen route has the minimum possible cost.

- Travel time: Time required for delivery, if travel-time data is available.
- Reliability: Ability to respond to unavailable routes.
- Computational efficiency: Time and memory required to find the route.

Analysis of possible routes

Using the weights shown in the graph, the following routes can be compared.

Route 1: S → A → B → D → G

Total cost = 1 + 3 + 3 + 3 = 10

Route 2: S → A → C → D → G

Total cost = 1 + 1 + 1 + 3 = 6

Route 3: S → A → C → G

Total cost = 1 + 1 + 2 = 4

Route 4: S → G

Total cost = 12

Comparison of route costs

Route	Total cost
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	4
S → G	12

Result: The minimum-cost route is S → A → C → G, with a total cost of 4.

Analysis: Although the direct route from S to G uses only one connection, its cost is 12. The route through A and C has three connections but costs only 4. Therefore, the agent should choose the route with the lowest cumulative cost, not necessarily the route with the fewest connections.

5. What sort of agent architecture is most suitable for this agent? Why?

Answer:

A goal-based agent using cost-based search is the most suitable architecture for this delivery problem.

A goal-based agent selects actions based on a specified objective. Here, the objective is to reach destination G from starting location S at the minimum possible cost.

Reasons for selecting this architecture

1. Goal-oriented planning: The agent knows that it must reach G.
2. Evaluation of alternatives: It considers different routes before selecting one.
3. Cost optimization: It calculates and compares cumulative route costs.
4. Optimal decision-making: It can select the minimum-cost route using a suitable search algorithm.
5. Replanning capability: It can search for another route if the network changes.

Suitable search algorithm: Uniform Cost Search (UCS)

Uniform Cost Search expands the path with the lowest cumulative cost first. It is suitable for this problem because the edges have different weights.

For the given graph, UCS identifies the route:

S → A → C → G

Total cost = 1 + 1 + 2 = 4

Analysis: A simple reflex agent would be less suitable because choosing an action based only on the current location may not produce the cheapest complete route. A goal-based agent using UCS considers cumulative costs and can find the optimal route when edge costs are non-negative.

Overall Conclusion

The delivery network illustrates how an intelligent agent can solve a route-planning problem through perception, action selection, performance evaluation, and search.

The analysis shows that the best route is S → A → C → G, with a minimum total cost of 4. A goal-based agent using Uniform Cost Search is suitable because it compares alternative paths and identifies the least-cost route. This approach improves delivery efficiency by avoiding unnecessarily expensive routes.